DSA06 DATA HANDLING AND VISUALIZATION | Activity: Correlation Detective | Slot C

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Time: 20 to 25 minutes Total: 20 marks Tool: R (ggplot2 / base R). Work alone; your case file is different from your neighbour's. Write your answers and paste your plots and code.

Case File CD-C01: The Online store Case [20 marks]

Briefing: You are the data detective hired by the operations manager of an online store. The file holds 16 weekly records from the festive quarter, with seven variables. Some relationships can be trusted. Others are traps: a hidden driver, a curve that fools r, and a single wrong record. Find them.

R name	What it measures	R name	What it measures
festive_index	Festive-season index (0-100)	complaints	Customer complaints
ad_clicks	Advert clicks (hundreds)	staff_online	Support staff online
parcels	Parcels delivered (hundreds)	returns	Return requests
delay_dev	Delivery time minus promised time (hours)		

Evidence: run this code in R to create your case file (do not edit it).

```
set.seed(59494)
n <- 16
festive_index <- round(rnorm(n, 57.1, 15.0), 1)
ad_clicks <- round(83.0 + 20.0 * (0.92 * (festive_index - 57.1) / 15.0 + 0.39 * rnorm(n)), 1)
parcels <- round(62.2 + 15.0 * (0.85 * (festive_index - 57.1) / 15.0 + 0.53 * rnorm(n)), 1)
delay_dev <- round(seq(-21.5, 21.5, length.out = n)[order(runif(n))], 1)
complaints <- round(10.4 + 30.2 * (delay_dev / 21.5)^2 + rnorm(n, 0, 1.0), 0)
staff_online <- round(rnorm(n, 16, 4.0), 0)
returns <- round(26 + 8.0 * (0.25 * (staff_online - 16) / 4.0 + 0.97 * rnorm(n)), 0)
staff_online[7] <- 29; returns[7] <- 51   # row 7 was typed in by a trainee
case <- data.frame(festive_index, ad_clicks, parcels, delay_dev, complaints, staff_online, returns)
```

Tasks:

1. The matrix. Use cor(case) to get the 7 x 7 correlation matrix (round to 2 decimals). State n and how many distinct pairs of variables there are. Draw a correlogram of the matrix (ggplot2 geom_tile with geom_text, or corrplot). [4]
2. The strongest clues. Name the strongest positive pair and the strongest negative pair (off-diagonal). Give r for each and call it weak (below 0.3), moderate (0.3 to 0.7) or strong (above 0.7). [3]
3. The deceptive zero. Plot complaints against delay_dev with a straight line, then add a quadratic fit (formula = y ~ x + I(x^2)). Report r and the R-squared of the quadratic fit. In one sentence, say why r misses this relationship. [4]
4. The outlier alibi. Plot returns against staff_online. Find the row that looks wrong and give its row number. Report r for all rows and r without that row, and say what the single record did to r. [4]
5. The prime suspect. ad_clicks and parcels are strongly linked. Report r(ad_clicks, parcels), then r(festive_index, ad_clicks) and r(festive_index, parcels). Compute the partial correlation of ad_clicks and parcels controlling for festive_index: correlate resid(lm(ad_clicks ~ festive_index)) with resid(lm(parcels ~ festive_index)). Write a two-sentence verdict: does ad_clicks cause parcels? Name one study design that could test causation. [5]

Detective's rule: *correlation measures a straight-line link only, it can be inflated by one point, and it never proves cause.*

Code of Conduct: I certify that this work is my own. Signature: _____

Activity - 7

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1.

A) The Correlation matrix shows the strength and direction of linear relationships between the seven variables.

```
Cor_matrix <- Cor(case)
round (Cor_matrix, 2)
```

The no. of observations is n = 16.

The no of distinct variable pairs is:

$$\frac{7(7-1)}{2} = 21$$

To draw Correlation Heat map:

```
library (ggplot2)
library (corrplot)

Corrplot ( Cor_matrix , method= "color", add Coef.Col= "black")
```

2.

A)

```
Cor.matrix <- Cor(case)
Cor_matrix [upper.tri(cor. matrix)]
```

The strongest pair is the pair which with the highest Positive Correlation coefficient r. The strongest negative pair with the lowest negative coefficient.

- Weak Correlation : $|r| < 0.5$
- Moderate Correlation : $0.5 \leq |r| \leq 0.7$
- Strong Correlation : $|r| > 0.7$

- Strongest positive pair : r = 0.94
- Strongest negative pair : r = -0.60

3)

1)

```
plot(case$delay_dev, case$Complaints,
     main = "Delivery Delay vs Complaints",
     xlab = "Delivery Delay (hours)",
     ylab = "Complaints")

abline(lm(Complaints ~ delay_dev, data = case),
       col = "red")

fit <- lm(Complaints ~ delay_dev + I(delay_dev^2), data = case)

Summary(fit)
cor(case$delay_dev, case$complaints)
Summary(fit)$r.squared
```

Interpretation: The relationship is curved rather than purely linear. Pearson's correlation coefficient measures only a straight line relationship, so it may be low even when a strong nonlinear relationships exists. The quadratic model captures the curved pattern.

4.)

A)

```
Plot( case$ staff_online , case$ returns,
      main = "return Vs Staff Online",
      Xlab = "Staff Online",
      Ylab = "Returns")

Cor (Case$ staff_online, case$returns)

case2 <- case [-7, ]

Cor (case2$staff_online. case2$returns)
```

Interpretation: Row 7 contains an intentionally altered record: staff_online = 29 and returns = 51.

Compare the correlation using all rows with the correlation after removing row 7. This identifies how one unusual record can influence the correlation coefficient.

5)

A)

```
Cor (Case$ ad_clicks, case$ parcels)
Cor( case$ festive_index , case$ad_clicks)
Cor( case$ festive_index, case$ parcels)

r_partial <- cor(
   resid (lm (ad_clicks) ~ festive_index , data=case)),
   resid ( lm ( parcels ~ festive_index, data= case ))
  )
  r_partial.
```

Verdict: Advertising clicks and parcels may be correlated, but correlation alone does not prove that advertising clicks cause more parcels. Festive-season activity may influence both variables. A randomized controlled experiment, where advertising exposure is randomly assigned, could help test causation.